How to Multiply

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1 Introduction

What is a number?
The answer to this question is not obvious, and we will dissect a set-theoretic approach to constructing numbers in this paper. We will assume nothing, no knowledge of numbers, no understanding of how numbers work. We will only define operations, and define numbers in ways that satisfy our intuitive understanding. From this we will show one fundamental idea: that all multiplication is, in an entirely rigorous sense, *addition*.

2 Constructing the Natural Numbers

We have no numbers in the beginning. Even more so, we have almost no sets at all. Thus, we can begin by looking at the only set we have: the empty set. We often denote the empty set by $\{\} = \emptyset$. With this tiny fraction of information, we will begin by making our first naming choice and assert that $0 = \emptyset$.

One of the core properties of the numbers we desire to find here is a sense of ordering. We want to find a rigorous way to make large claims, like $0 < 1$. We will proceed in an incredibly clever way, exchanging $<$ for one of the core set-theoretic relations: $\in$. Since we have just discovered that all numbers are actually sets, $0 < 1$ *actually* means $0 \in 1$. That is, $0 < 1 \iff 0 \in 1$.

But what *is* 1? The only concrete piece of information we know about 1 is that 0 must be contained in 1. Motivated by this idea of an ordering, we will cleverly proceed by defining the *successor function* of a set x:

$$S(x) = x \cup \{x\} = x^+ = \text{“}x + 1\text{”}$$

In an intuitive sense, since $\{x\}$ is in the union of $S(x)$, $x \in S(x)$. Additionally, if x is a number with the property that every number “before” x is an element of x, then every number smaller than x will *also* be an element of $S(x)$, since every element of x is contained within $S(x)$.
As an example, we will construct the first few natural numbers:

$$\begin{aligned}
0 &= \{\} = \emptyset \\
1 &= \{0\} = \{\{\}\} \\
2 &= \{0, 1\} = \{\{\}, \{\{\}\}\} \\
3 &= \{0, 1, 2\} \\
&\vdots
\end{aligned}$$

Well, what happens when we hit an "end"? This set is the first "limit ordinal," or a number (ordinal) that is not the successor of any other set, and is often denoted by ω.
That is,

$$\omega = \{0, 1, 2, 3, \dots\} = \mathbb{N}$$

This is why the natural numbers contain zero! When taking a set-theoretic approach, it only makes sense for ω to be the fundamental *definition* of the natural numbers.

Additionally, there is no real reason for us to stop here. We can continue by taking the successor of ω, leading us to work with ordinals and ordinal arithmetic. However, the scope of this paper will stay inside numbers we are generally familiar with.

2.1 Adding Natural Numbers

We will define the binary operator $+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ *recursively* based on the successor function—the only function we have. This is a trend that will continue, as every operator we construct will rely completely on tools already created. Specifically, given arbitrary natural numbers x and y, $+$ is defined as follows:

- $x + 0 = x$
- $x + y^+ = (x + y)^+$

This is pretty incredible for a couple of reasons. For one, addition is not necessarily commutative!! Of course it happens to be, but the definition in no way directly implies that $x + y = y + x$. In fact, for larger ordinals (particularly limit ordinals!) addition is actually *not* commutative. Fascinating!

If you are not yet fully convinced by this definition, I encourage you to try a couple of small examples! As an exercise, I encourage you to prove that addition is associative—that is, $(x+y)+z = x + (y + z)$ for all natural numbers x, y, z. (Hint: use induction!)

2.2 Multiplying Natural Numbers

Similarly, we will define the binary operator $\cdot : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ recursively based on addition. This form is nearly identical to that of addition, with this operation naturally proceeding addition. Given arbitrary natural numbers x and y, $\cdot$ is defined as follows:

- $x \cdot 0 = 0$
- $x \cdot y^+ = x \cdot y + x$

Similarly to addition, multiplication is also not necessarily commutative!! While multiplication does happen to be commutative in the natural numbers (as we would hope), limit ordinal multiplication ends up being *very* far commutative. That will not be covered here, but I recommend you check it out! It's very cool.

2.3 Concluding Thoughts

Commutativity of addition and multiplication in $\mathbb{N}$, along with other common statements in $\mathbb{N}$, will be used frequently in this paper. Although we will not show these results here, all of them can proven—try it!

Once you come back, one big result is that multiplication in the natural numbers is actually just repeated addition. Well, who cares? Is this not a result that we took to be truth in about 3rd grade? How is this different?

Well, what what about numbers that aren't natural numbers? What comes next?

3 Constructing the Integers

The key idea that defines the set of integers, $\mathbb{Z}$, is the introduction of negative numbers. But what does this actually mean? If we were to continue thinking about numbers as sets, what would a negative number look like, especially if we wanted the above definitions of succession, addition, and multiplication to hold?

We will substitute thinking about negation for thinking about solutions to the system $x + m = n$, with $m, n \in \mathbb{N}$. This turns out to be incredibly clever! In the mathematical systems we are used to, a solution would give $x = n - m$. (We will call this our "informal" understanding.) However, without any sense of subtraction, this solution doesn't make any sense. Thus, instead of thinking about $n - m$, we will instead choose to consider the ordered pair (n, m), where (n, m) "represents" what we want $n - m$ to be.

Well, what's an ordered pair? Since everything is a set, we will define the arbitrary ordered pair (a, b) to be the set $\{a, \{a, b\}\}$. (I encourage you to think about why!)

However, we immediately have another problem. If every integer can be represented by the ordered pair $(m,\, n)$, where $(m,\, n)$ is analogous to $m - n$ in the integers we know, then doesn't every integer have infinitely many values of m and n that satisfy?

Yes! For instance, solutions to the equation $x + 4 = 2$ can be thought of as both $(2, 4)$ and $(3, 5)$. Our solution to this problem will be define an *equivalence relation* that will create an equivalence class. Thus, every integer is not a single ordered pair, but a *class* of ordered pairs.

Let us begin considering two ordered pairs of natural numbers: $(a,\, b)$ and $(c,\, d)$. If we wanted it to be the case that $(a,\, b) \simeq (c,\, d)$, then this should be equivalent to our informal notion of what $a - b = c - d$ means. However, since we cannot subtract in $\mathbb{N}$, we will instead formally define the following relation:

$$(a,\, b) \simeq (c,\, d) \iff a + d = b + c$$

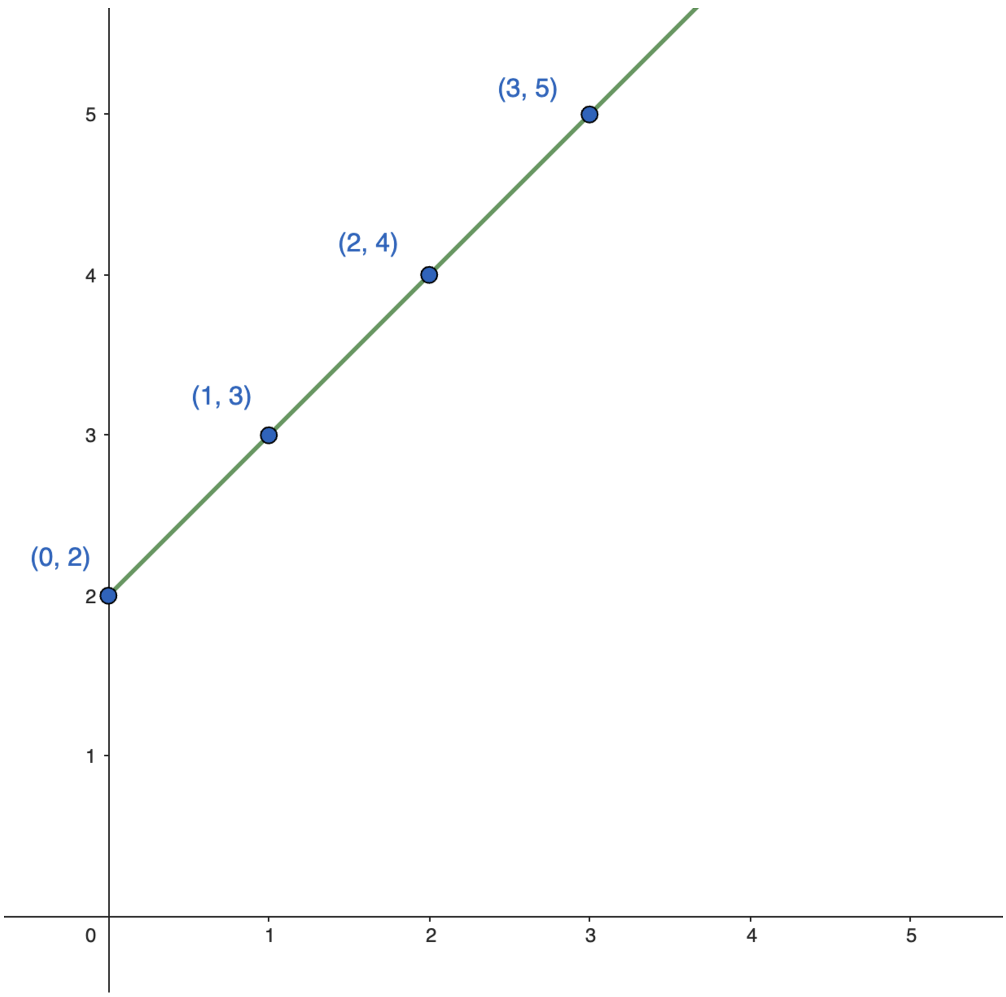
We will continue by asserting that $\simeq$ is an equivalence relation. The proof of this is enlightening and left to the reader, much of which relies on the fact that addition is commutative.

We can now write our solution to the initial example in the following notation:

$$[(0,\, 2)] = \{(0,\, 2),\ (1,\, 3),\ \ldots,\ (n,\, n+2),\ \ldots\} \qquad n \in \mathbb{N}$$

We note that $[(0,\, 2)]$ is just one possible representation of this equivalence class—writing $[(1,\, 3)]$ would refer to the same equivalence class.

One particularly interesting way of thinking about this idea is to consider a geometric interpretation of these equivalence classes. We say that two points $(a,\, b)$, $(c,\, d) \in \mathbb{N} \times \mathbb{N}$, are equivalent if and only if $a+d = b+c$. Returning to an intuitive sense of $\mathbb{Z}$, we can rewrite this as $a-b = c-d \implies \frac{a-b}{c-d} = 1$. Thus, we can visualize these points as being the lattice points lying on a line with a slope of 1 passing through any given representative of out equivalence class:



If we define the representative of a given equivalence class to be $(0,\, k)$ or $(k,\, 0)$ (the proof of the existence of this representative is left to the reader, though the above diagram should give some intuition for why this is the case), then every integer can be thought of as the *projection* of all $(a,\, b) \in \mathbb{N} \times \mathbb{N}$ onto the x or y-axis along a line with a slope of 1. This uniquely maps every lattice point in the first quadrant onto a single integer. Woah!

We can now finally officially define $\mathbb{Z} = (\mathbb{N} \times \mathbb{N})/ \simeq$. This is read as "$\mathbb{Z}$ is the set of all ordered pairs $(a,\, b)$ where $a, b \in \mathbb{N}$ under the equivalence relation $\simeq$." We note carefully that elements in $\mathbb{N}$ are *not* elements of $\mathbb{Z}$. We currently have defined $1 \in \mathbb{N}$ and $1 \in \mathbb{Z}$ *entirely differently.*

3.1 Adding Integers

We will define addition in the integers using the only knowledge of addition that we already have—adding in the natural numbers. To reiterate: we began with the successor function $S(x)$, and every function we have created since has been defined through the use of *previously existing* functions.

However, we will again carefully note that + in $\mathbb{N}$ and + in $\mathbb{Z}$ are completely different functions. We probably should give them different names, but I find this notation to be unambiguous as the correct + relies only on the type of input. Given $[(a,\, b)],\ [(c,\, d)] \in \mathbb{Z}$, define

$$[(a,\, b)] + [(c,\, d)] = [(a + c,\, b + d)]$$

For intuition behind this form, we recall that $(a,\, b)$ in our informal sense is "the same" as $a - b$. If we were adding $a - b$ and $c - d$, and we grouped together the "positive" terms and "negative" terms, we would arrive at $(a + c) - (b + d)$. Returning to our rigorous definition of an integer, we then arrive at the desired form of $[(a + c,\, b + d)]$.

3.2 Multiplying integers

Extending multiplication to the integers from the natural numbers will be virtually identical to extending addition to the integers from the natural numbers as we performed above. We will define $\cdot : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ as follows:

$$[(a,\, b)] \cdot [(c,\, d)] = [(ac + bd,\, bc + ad)]$$

Although this form looks slightly more complicated, the intuition behind how we arrive at this definition is the same as how we considered our definition for addition. We will again informally consider multiplying two integers $a - b$ and $c - d$. Upon grouping by sign, we arrive at $(ac + bd) - (bc + ad)$, which rigorously translates to our above desired form of $[(ac + bd,\, bc + ad)]$

3.3 Concluding Thoughts

Here, we find that multiplication in the integers is actually just multiplication and addition in the *natural numbers*! And since multiplication in $\mathbb{N}$ is repeated addition in $\mathbb{N}$, we can conclude that multiplication in $\mathbb{Z}$ is just repeated addition!

If you wish, you can prove all of the statements we know to be true about integers—that multiplication and addition are well-defined (consistent regardless of which element you use from an equivalence class), that commutativity and associativity hold, that multiplication has the cancellation property, that subtraction exists, etc. All of these proofs are enlightening, and I urge you to pick a couple that interest you and give them a try.

One last idea of note: one may be disturbed by the idea that $\mathbb{N} \not\subseteq \mathbb{Z}$. Our solution will be to say that the non-negative integers have the same *structure* as the natural numbers. (Both are called a Peano system, and all Peano systems are essential the same system.) Thus, if we are working not concerned about the foundations of numbers (as we are here), we will choose to say that $\mathbb{N} \subseteq \mathbb{Z}$ and justify this by saying that there exists an *isomorphism* between $\{0, 1, 2, ...\}$ and $\{[(0,\, 0)],\, [(1,\, 0)],\, \ldots,\, [(n, 0)],\, \ldots\} \quad n \in \mathbb{N}$.

Well, what about solutions to equations like $2x + 1 = 5$? x certainly takes no solution in $\mathbb{Z}$, how could we possibly solve that?

4 Constructing the Rational Numbers

This construction will be very similar to the construction of the integers, as we already have an informal sense of what we want a rational number to look like. We want a rational number to be of the form $\frac{a}{b}$, where $a \in \mathbb{Z}$, $b \in \mathbb{Z}\backslash\{0\}$. However, we will again run into the same problem of having too many *different* rationals if we try to create $\mathbb{Q}$ through the ordered pair of integers (a, b) with $b \neq 0$. (For example, we want $\frac{1}{3}$ to be the same number as $\frac{2}{6}$.)

Our solution? Equivalence classes! Let us define the following relation $\simeq$ on $\mathbb{Q}$. Given $(a, b), (c, d) \in \mathbb{Z} \times \mathbb{Z}\backslash\{0\}$, we will define

$$(a, b) \simeq (c, d) \iff ad = bc$$

Returning to our informal sense of rationals in search of intuition, writing $\frac{a}{b} = \frac{c}{d}$ is the same as writing $ad = bc$. Since we have no sense of division, we choose to take the second form for our above relation. We will accept without proof that this relation is an equivalence relation, though the proof is instructive and left as an exercise.

Just as we defined $\mathbb{Z}$ through an ordered pair of $\mathbb{N}$ and an equivalence relation, we can now define $\mathbb{Q} = (\mathbb{Z} \times \mathbb{Z}\backslash\{0\})/\simeq$. As an example, the equivalence class

$$[(1, 1)] = [(2, 2)] = \{\ldots, (-2, -2), (-1, 1), (1, 1), (2, 2), \ldots\}$$

can be intuitively thought of as 1, while

$$[(1, 3)] = [(17, 51)] = \{\ldots, (-2, -6), (-1, -3), (1, 3), (2, 6), \ldots\}$$

can be intuitively thought of as $\frac{1}{3}$.

We can also consider a geometric interpretation of the rational numbers! We say that two points $(a, b), (c, d) \in \mathbb{Z} \times \mathbb{Z}\backslash\{0\}$ are equivalent if and only if $ad = bc$. Returning to our intuitive notion of $\mathbb{Q}$, we can rewrite this as $\frac{a}{b} = \frac{c}{d} = \frac{a-c}{b-d}$. Placing a hole at $(0, 0)$ and flipping the x and the y axis, we can visualize these points as being the lattice points lying on a line with a slope of $\frac{a}{b}$ passing through the origin!

We find that every lattice point (except for ones with a y-coordinate of 0) is uniquely mapped onto a line with a slope of $\frac{a}{b}$! (The "slope" here being the change in x rather than the change in y.)

4.1 Adding Rational Numbers

This construction will be nearly identical to our construction of addition in $\mathbb{Z}$. We will use our intuitive sense of $\mathbb{Q}$ to create a definition of addition that matches our informal understanding of what addition of rational numbers *should* be.

Let $[(a, b)], [(c, d)] \in \mathbb{Q}$. We define:

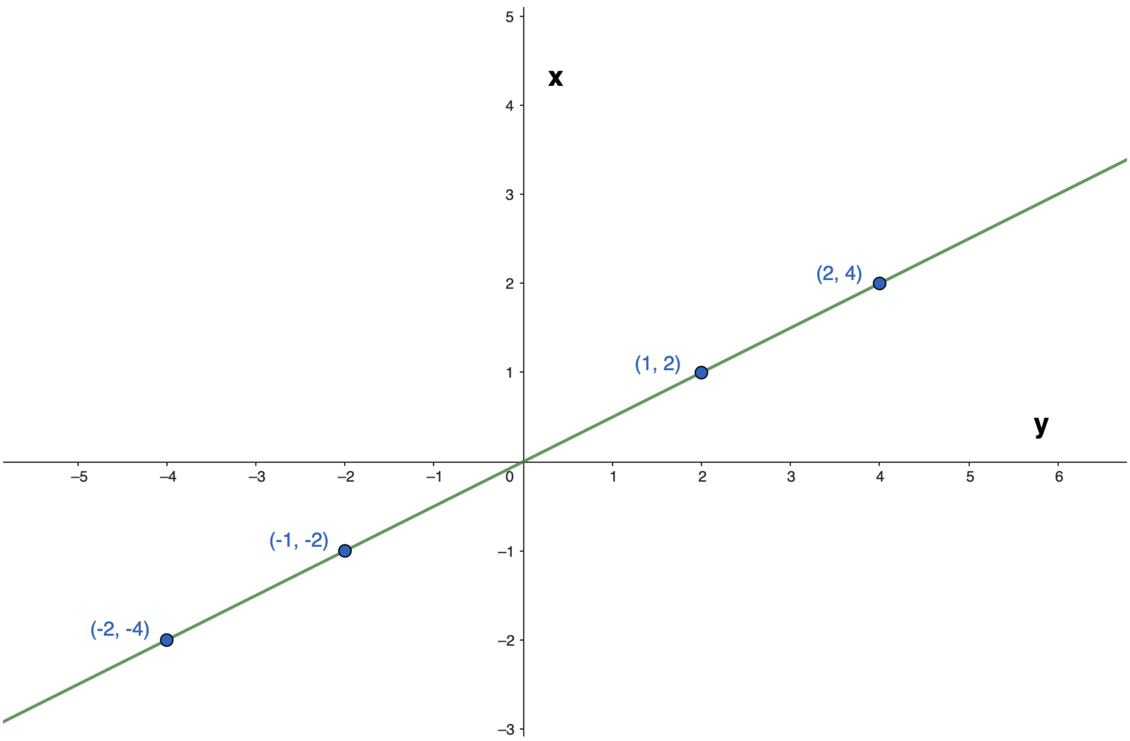
$$[(a, b)] + [(c, d)] = [(ad + bc, bd)]$$

This aligns with our intuitive notion of addition in the rationals, as $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$.

4.2 Multiplying Rational Numbers

Following a similar construction, we define:

$$[(a, b)] \cdot [(c, d)] = [(ac, bd)]$$



This result comes from our informal notion that $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$

We again will assume that both multiplication and addition are well-defined and contain all of the properties we expect. These proofs will again be left as exercises to the curious reader.

4.3 Concluding Thoughts

Similarly to before, we find that both addition and multiplication in the rationals rely on multiplication and addition in the integers. (And so on.) Thus, so far, all multiplication is still just addition!

Additionally, we again find that $\mathbb{Z} \not\subseteq \mathbb{Q}$. However, we find that the "integers in the rationals," or any equivalence class in the rationals containing $(n, 1)$, is again *isomorphic* to the $\mathbb{Z}$ that we are now somewhat familiar with. Therefore, when working in an environment that doesn't care about how any of these sets are formed, it is perfectly reasonable to state that $\mathbb{Z} \in \mathbb{Q}$.

For one final thought, let us regain a sense of scope and remember where we began. As an example, let us consider what the rational number 3 *really* looks like. In the rationals, 3 is defined as $[(3, 1)]$. Expanding this notation:

$$3 = [(3, 1)] = \{\dots, (-6, -2), (-3, -1), (3, 1), (6, 2), \dots\}$$

But each element of this equivalence class is an ordered pair of integers:

$$3 = \{\dots, \{-3, \{-3, -1\}\}, \{3, \{3, 1\}\}, \dots\}$$

And expanding just the first integer (I will now underline natural numbers):

$$3 = \{\dots, \{\{(\underline{0}, \underline{3}), (\underline{1}, \underline{4}), \dots\}, \{-3, -1\}\}, \{3, \{3, 1\}\}, \dots\}$$

And expanding just the first ordered pair of natural numbers:

$3 = \{\ldots, \{\{\{\{\emptyset\}, \{\emptyset, \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}, \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}\}\}, (\underline{1}, \underline{4}), \ldots\}, \{-3, -1\}\}, \{3, \{3, 1\}\}, \ldots\}$

Seemingly simple questions like "What is 3?" have suddenly become *enormously* complicated. Good thing no number system is more complex than the rationals!

5 Constructing the Real Numbers

We often notate real numbers as successive approximations. For example, we write $e = 2.71828\ldots$ to mean that e is 2.71828—at least to 5 decimal places. We could easily write $e = 2.718281828$ for a better approximation. If one were to continue writing down the digits of e, you would get "closer" to the "true value" of e. This intuitive notation is not far off from how we will define a real number. Differing from the sets we have worked with so far, one representation of a real number will not be an ordered pair of rational numbers but a *sequence* of rational numbers. Thus, we can write

$$e = 2,\ 2.7,\ 2.71,\ 2.718,\ \ldots$$

However, we run into the same problem we always have! There are simply too many representations of every real number. For example, we can also write e as the following sequence of rational numbers:

$$e = 2,\ 3,\ \frac{8}{3},\ \frac{11}{4},\ \frac{19}{7},\ \frac{87}{32},\ \ldots$$

Or as the following:

$$e = 1717,\ -34890,\ 500,\ 0,\ 5,\ 2.7,\ 2.718281828, \ldots$$

Even though the beginning started sporadically, the "tail" still tends towards e. As it turns out, this tail-end behavior is all we care about!

5.1 Cauchy Sequences

Every sequence of rational numbers is actually a function from $\mathbb{N}$ into $\mathbb{Q}$. Intuitively, we "assign" every natural number a rational number. Each sequence will be of the form $\{a_0, a_1,\ a_2,\ ...\}$, which we will shorthand to $(a_n)_0^\infty$ and further shorthand to (a_n).

One useful sequence to take note of is any sequence that tends towards 0. If some sequence (a_n) tends to 0, then we say that for any small $\epsilon > 0$ there exists some natural k such that $a_m < \epsilon \quad \forall m > k$. This behavior of tending towards a value is *convergence*. For example, we say that the sequence $(1/n)$ converges to 0, as for any $\epsilon > 0$ we can choose $N > 1/\epsilon$ which will imply $1/N < \epsilon$.

How about sequences that certainly converge, but not to a rational number? For example, 2, 2.7, 2.718, ... definitely is *not* converging to a rational number. Despite this, it definitely seems like its getting closer to something. We call this type of sequence a *Cauchy sequence*. Formally, we say that a sequence (a_n) is a Cauchy sequence if the difference between terms approaches 0. That is, $\forall \epsilon > 0\ \exists k\,(|a_m - a_{m+1}| < \epsilon \quad \forall m > k)$.

5.2 Equivalence Relations

Now to solve the original problem: an abundance of Cauchy sequences that we want to represent to the same real number. Let us define $\mathfrak{C}_{\mathbb{Q}}$ to be the set of all Cauchy sequences of rational numbers. Given two sequences (a_n), $(b_n) \in \mathfrak{C}_{\mathbb{Q}}$, we will define the following relation:

$$(a_n) \simeq (b_n) \iff a_n - b_n \text{ tends towards } 0$$

The proof that $\simeq$ is an equivalence relation is interesting, and the proof that transitivity holds is particularly clever. While these proofs will not be conducted here, they are well worth a moment of thought! Now, we can finally define $\mathbb{R} = (\mathfrak{C}_{\mathbb{Q}})/\simeq$.

This is the true reason why $0.999\cdots = 1$. $0.999\ldots$ doesn't equal 1 based on declaring that $S = 0.999\ldots$ and continuing algebraically, nor does $0.999\ldots$ really equal 1 by considering a convering series. $0.999\cdots = 1$ because the difference between *every* Cauchy sequence of $0.999\ldots$ and *every* Cauchy sequence of 1 *converges to zero*.

5.3 Arithmetic in the Real Numbers

Multiplication and addition are surprisingly (or maybe not so surprisingly) simple in $\mathbb{R}$! Given real numbers $[(a_n)]$, $[(b_n)]$, we define:

$$[(a_n)] + [(b_n)] = [(a_n + b_n)]$$

$$[(a_n)] \cdot [(b_n)] = [(a_n \cdot b_n)]$$

Despite how simple this form looks, proving that addition and multiplication are well-defined is undoubtedly more complicated here than in the sets we have already worked with. This is worth working out, and my hint is to recall your approach to proving that $\simeq$ is transitive.

5.4 Concluding Thoughts

The defining factor that separates the real numbers is that the real numbers satisfy the Completeness Axiom. In an intuitive sense, this means that any non-empty unbounded subset of $\mathbb{R}$ contains a bound *within* $\mathbb{R}$. This is not true in a set like $\mathbb{Q}$. For example, the set of all rational numbers less than the positive solution to $x^2 = 2$ *does not* contain an upper bound within $\mathbb{Q}$, while the set of all real numbers less than the positive solution to $x^2 = 2$ *does* contain an upper bound within $\mathbb{R}$. (We call this $\sqrt{2}$.)

So, in the end, every multiplication problem of two real numbers is actually just a sequence of multiplication of two rational numbers which is just two sets of multiplication of integers which is multiplication and addition of natural numbers. But since multiplication of natural numbers is just repeated addition of natural numbers, all multiplication is addition!

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